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MusicLIME: Explainable Multimodal Music Understanding

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***Abstract*—Multimodal models are critical for music understanding tasks, as they capture the complex interplay between audio and lyrics. However, as these models become more prevalent, the need for explainability grows—understanding how these systems make decisions is vital for ensuring fairness, reducing bias, and fostering trust. In this paper, we introduce MUSICLIME, a model-agnostic feature importance explanation method designed for multimodal music models. Unlike traditional unimodal methods, which analyze each modality separately without considering the interaction between them, often leading to incomplete or misleading explanations, MUSICLIME reveals how audio and lyrical features interact and contribute to predictions, providing a holistic view of the model's decision-making. Additionally, we enhance local explanations by aggregating them into global explanations, giving users a broader perspective of model behavior. Through this work, we contribute to improving the interpretability of multimodal music models, empowering users to make informed choices, and fostering more equitable, fair, and transparent music understanding systems.**

***Index Terms*—Explainable Artificial Intelligence, Music Understanding, Multimodality**

I. INTRODUCTION

As artificial intelligence (AI) continues to evolve, researchers are increasingly focusing on multimodal approaches to harness the strengths of deep learning (DL) across diverse types of data. These multimodal models integrate various data sources, such as text, audio, and images, to enhance accuracy and make better use of the available data [1]. In the Music Information Retrieval (MIR) domain, multimodal approaches are becoming increasingly prominent as they combine audio and lyrical data to achieve more precise music analysis [2]. This includes tasks such as mood classification [3], emotion recognition [4], and music auto-tagging [5]. However, the complexity of multimodal models amplifies transparency challenges. The interaction between modalities makes understanding their decisions harder, adding to the transparency issues already present in unimodal systems. This lack of interpretability can obscure the decision-making process, impacting the reliability and fairness of the models.

Explainable AI (XAI) has emerged as a crucial area of research focused on enhancing the interpretability and transparency of machine learning models [6], [7]. XAI methods are essential for understanding how models make decisions and the underlying data [8], thereby improving user trust and facilitating responsible AI deployment [9], [10]. Among these methods, Local Interpretable Model-agnostic Explanations (LIME) stands out as a seminal and widely accepted approach in the XAI field [11]. It provides local explanations by systematically perturbing input features and observing how predictions change, offering a valuable tool for examining model behavior at the instance level. Recent advances in the area include AUDIOLIME, a variant of LIME adapted specifically for the audio domain, which applies the same principle to audio-specific features [12]. In the music domain, XAI methods have been applied to interpret models through attention mechanisms [13], perceptual musical features [14], genre and spectral prototypes [15], [16], and concept-based explanations [17].

While existing XAI methods have advanced explainability in the music domain, there is a notable gap in approaches tailored to multimodal models, particularly in music, which combines both audio and lyrical data. Multimodal explainability offers a significant advantage over unimodal methods by providing a more comprehensive understanding of how different modalities interact within a model's decision-making process. Unlike unimodal approaches, which analyze each modality in isolation and can lead to incomplete or misleading insights, multimodal explanations enable a holistic overview of the model's behavior by revealing the contributions and interactions between features from different modalities. This allows users to better understand the intricate dynamics between, for example, lyrical content and audio features in music. Several studies have explored XAI methodologies for multimodal settings [18], including the development of a multimodal LIME approach for image and text sentiment classification [19]. However, these methodologies have yet to be fully applied to the MIR domain, leaving a gap in explainability for multimodal music models.

In this paper, we introduce MUSICLIME, a model-agnostic feature importance explanation method specifically designed for multimodal music understanding systems. As part of our methodology, we curated two datasets tailoring them for multimodal music emotion and genre recognition and developed a transformer-based multimodal model to tackle these challenges. MUSICLIME addresses the challenge of explaining the interactions between audio and lyrical features, providing a more comprehensive view of how these modalities contribute to predictions. Additionally, we provide global explanations by aggregating local explanations, offering a broader understanding of the model's overall behavior. All code, implementation details, and instructions for reproducing the results are available in our GitHub repository [1].

II. METHODOLOGY

A. Model Architecture

We experimented with two modalities: text (lyrics) and audio, utilizing a language model for text and an audio model respectively. These two transformer-based models were combined into a single multimodal model by concatenating their embeddings into a unified vector, which is then fed into a classification head. The aim was to establish a baseline model that will be used to evaluate the effectiveness of our MUSICLIME method in providing insights into the model's behavior across music genre and emotion classification tasks. Notably, our approach can be effortlessly adapted to any model of choice.

After a thorough investigation of model architectures, we choose to experiment with ROBERTA [20] as our language model and Audio Spectrogram Transformer (AST) [21] as our audio model. These models were chosen for their ease of implementation and

[1] https://github.com/IamTheo2000/MusicLIME

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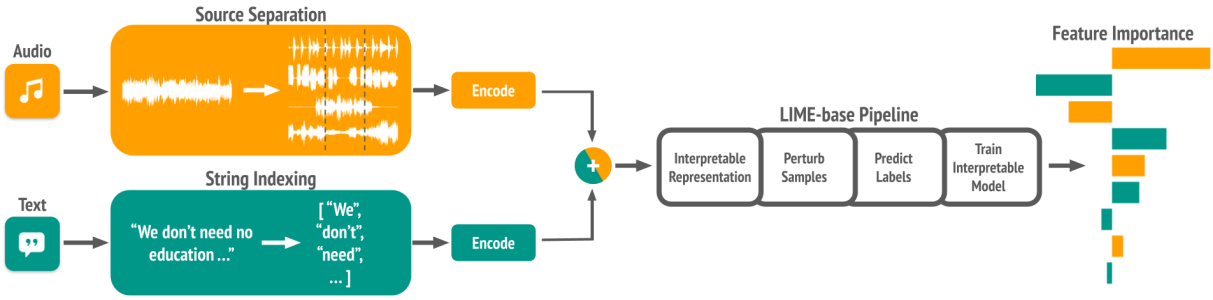

Fig. 1: Overview of MUSICLIME

their balanced size and performance. It is important to note that our methodology is model-agnostic and can be easily applied to larger models. We utilized ROBERTA-LARGE [2], by first preprocessing and tokenizing the input lyrics. We generated mel-spectrograms from the audios, using the provided FEATURE EXTRACTOR, specific for the AST [3]. Our multimodal framework was created by concatenating the pooled output from the AST with the CLS token from ROBERTA. This combined feature vector was then fed into a classification head, comprising a normalization layer and a fully connected layer, to generate the final predictions. Fine-tuning on our dataset was performed both separately for each modality and jointly for the multimodal setting.

B. Unimodal and Multimodal Explainability

In this study, we selected LIME as the foundation for our explainability approach due to its simplicity, widespread adoption, and proven effectiveness in providing intuitive model explanations. LIME has been successfully adapted to various domains and modalities, including images, audio, and text. In the music domain, the two primary modalities of interest are text and audio. For text-based models, LIME assigns importance scores to individual words, indicating their contribution to the final prediction. In the audio domain, while spectrograms can be treated as images to highlight important parts using LIME, such explanations are often difficult to understand or interpret. A more suitable approach is AUDIOLIME, a specialized version that segments audio into meaningful time intervals and isolates components like vocals or instruments, resulting in more comprehensible and intuitive explanations.

While the aforementioned approaches provide useful explanations for unimodal models, the multimodal nature of music requires an adaptation that can capture the intricate interplay between its different modalities. To address this limitation, we created MUSICLIME, an extension of the LIME FRAMEWORK specifically tailored for multimodal music understanding models. MUSICLIME is designed to explain how both audio and lyrical features interact and contribute to a model's predictions, offering a more comprehensive view of the decision-making process in music classification tasks. An overview of MUSICLIME is shown in Figure 1. MUSICLIME processes the two modalities separately before integrating them. For the audio modality, our approach builds on AUDIOLIME by splitting the input into temporal segments and further decomposing each segment into its constituent sources. Additionally, our GitHub repository [4] offers the option to use Demucs, another highly regarded state-of-the-art source separation model [22]. Each audio instance is divided into 10 segments and split into the components: vocals, drums, bass, and other instruments. For the text modality, we follow an approach similar to traditional LIME for text, where the input is split into individual words. After pre-processing, the features from both modalities are encoded and concatenated into a unified feature vector, indicating the presence or absence of features. Following LIME's methodology, we perturb this vector representation by selectively including or excluding features, allowing us to observe changes in the model's predictions. Using these results, we train an interpretable linear model to approximate the multimodal model's behavior locally. This approach enables us to compute feature importance scores for both audio and text simultaneously, facilitating a direct comparison of their contributions to the model's decision-making process.

[2]https://huggingface.co/docs/transformers/en/model_doc/roberta

[3]https://huggingface.co/docs/transformers/en/model_doc/audio-spectrogram-transformer

[4]https://github.com/IamTheo2000/MusicLIME

C. Global Aggregations of Local Explanations

To gain a comprehensive understanding of the model's behavior beyond individual instances, generating class-wide explanations, we implemented Global Aggregations of Local Explanations as described in [23]. In our work, we apply two methods: (1) Global Average Importance, and (2) Global Homogeneity-Weighted Importance.

The Global Average Class Importance is calculated as follows:

$$I_{cj}^{\text{AVG}} = \frac{\sum_{i \in S_c} |W_{ij}|}{\sum_{i \in S_c : W_{ij} \neq 0} 1} \tag{1}$$

where S_c is the set of all instances classified as class c, and W_{ij} is the weight of feature j for instance i provided by LIME.

The second method involves calculating a normalization vector for each feature j across all classes L as $p_{cj} = \frac{\sqrt{\sum_{i \in S_c} |W_{ij}|}}{\sum_{c \in L} \sqrt{\sum_{i \in S_c} |W_{ij}|}}$. The normalized LIME importance p_{cj} represents the distribution of feature j's importance across classes. The Shannon entropy of this distribution is calculated as $H_j = -\sum_{c \in L} p_{cj} \log(p_{cj})$, measuring the homogeneity of feature importance across multiple classes. Finally, the Homogeneity-Weighted Importance is:

$$I_{cj}^{H} = \left(1 - \frac{H_j - H_{\min}}{H_{\max} - H_{\min}}\right) \sqrt{\sum_{i \in S_c} |W_{ij}|} \tag{2}$$

where $H_{\min}$ and $H_{\max}$ are the minimum and maximum entropy values across all features. Intuitively, this method penalizes features that influence multiple classes, whereas higher weights are assigned to features that are important for specific classes.

Implementing (1) and (2), we note that for multimodal models, homogeneity-weighted importance does not accurately capture the influence of multimodal features. This is due to the different nature of the features. While words are distinct, audio features encapsulate different sounds. For example, a vocal feature can contain various styles ranging from soothing singing to screams and shouts. As a result, the same audio features can impact many classes for different reasons. Since Homogeneity-weighted importance punishes features that impact multiple classes, lower weights are assigned to audio features compared to the text ones, which is inaccurate. Therefore, global average class importance is more suited for multimodal analysis.

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III. Experiments

A. Datasets

Although the Music Information Retrieval (MIR) community has created various multimodal datasets [24], many of which can be found on ISMIR's resource page[5], finding a dataset that includes both audio and lyrics remains challenging due to copyright restrictions. For this study, we curated two datasets: *Music4All* [25] (*M4A*), a multimodal dataset with both audio and lyrics, and a manually constructed multimodal subset of AudioSet [26], where we combined audio from AudioSet with lyrics sourced from external databases.

M4A provides 30-second audio clips and lyrics for each instance, along with metadata including genre labels and valence-energy values. Using these metadata, we categorized the songs into nine distinct genres (*heavy music*, *punk*, *rock*, *alternative rock*, *folk*, *pop*, *rhythm music*, *hip hop* and *electronic*) based on Musicmap [6] and nine distinct emotion categories derived from Russel's circumplex model [27]. Songs that did not fit into one of these nine categories, such as soundtrack music, were excluded from the final dataset. To ensure label accuracy, we cross-referenced the genre labels with Spotify's artist genre classifications, refining the dataset to include around 60,000 songs, with 50,000 reserved for training. We maintained a data split where no artist from the training and validation sets appeared in the test set, ensuring that the model was evaluated on truly unseen data for generalizability.

To further validate our methodology and ensure that our results are not dependent on a single dataset, we created a small multimodal dataset based on a subset of music-related recordings from AudioSet [26]. AudioSet contains descriptive labels (e.g., *fireworks*, *harmonica*) of YouTube videos' audio segments. We focused on music samples and matched the song titles with lyrics from two openly available sources[7][8]. This process involved fetching video titles for all entries, filtering out non-song instances (such as compilations, remixes, or series episodes), extracting artist and song names from the titles, and retrieving the corresponding lyrics when available. This procedure resulted in a set of 308 audio-lyrics pairs, which were used to evaluate the robustness of our approach across different datasets, thereby introducing a new curated multimodal music dataset.

B. Experimental Setup

Our configurations utilized NVIDIA's V100 and P100 GPUs, with 16 GB of RAM each. All models were implemented using the PyTorch framework, with additional utility libraries provided by Hugging Face. A preprocessing step was necessary for our data. For the textual data, this involved standard data-cleaning procedures, such as converting characters to lowercase and removing punctuation. After cleaning, the text was tokenized into sequences of up to 256 tokens. For the audio data, we extracted mel-spectrograms with 128 mel bands, utilizing a window and FFT size of 512, with a sampling rate of 44100 Hz. For the training of each model, default values for learning rates, batch sizes, and the number of epochs were utilized. A checkpointing mechanism was implemented throughout the training process to ensure that the model state corresponding to the highest validation accuracy was preserved.

To generate the global aggregates, we combined the weights produced by multiple instances, each generated with a different number of perturbations. Specifically, for the *M4A* dataset, we aggregated the results from 640 instances for the lyrical model, 240 instances for the audio model, and 128 instances for the multimodal model. For the *AudioSet* dataset, we combined the results of all the instances for the language model and 232 for the audio and multimodal models. The number of perturbations per instance for the language, audio, and multimodal models were 2500, 2000, and 5000 respectively. Finally, to visualize the aggregate weights of the words for each class and facilitate comparisons, we employed GloVe embeddings combined with t-SNE for dimensionality reduction.

[5]https://ismir.net/resources/datasets/

[6]https://musicmap.info/

[7]https://docs.genius.com/

[8]https://lyrics.lyricfind.com/

Model	Test Acc.
Lyrical Emotion (RoBERTa)	32.33%
Audio Emotion (AST)	48.29%
Multimodal Emotion	**48.53%**
Lyrical Genre (RoBERTa)	45.14%
Audio Genre (AST)	53.75%
Multimodal Genre	**57.34%**

TABLE I: Model performance summary

IV. Results & Discussion

Table I summarizes the performance of our models on the *M4A* dataset. Overall, the multimodal model consistently outperforms the unimodal models, demonstrating the value of combining text and audio features in music classification. The language model showed limited accuracy in predicting emotions but performed really well at identifying specific genres, such as *hip hop* and *heavy music*, likely due to recurring thematic elements in the lyrics, as further elucidated by our explanations (see Figures 2 and 3). Conversely, the audio model, generally outperformed the lyrical model across tasks, especially in emotion classification, indicating that mood-related information is more effectively captured in the audio domain. Additionally, genre prediction proved more accurate than emotion prediction, which may be attributed to the inherently subjective nature of human emotions [28] on one side, but also to the distinctive features of various genres, whether in lyrics (e.g., hip hop) or audio (e.g., vocals and drums in punk music). These observations are further validated by our multimodal explanations presented in the following paragraphs. Overall, the results emphasize the complementary strengths of each modality and highlight the importance of using multimodal explanations to better understand model behavior.

Figure 2 demonstrates the effectiveness of our approach and highlights its advantages over unimodal explanations. The Figure presents global multimodal explanations for *hip hop*, *punk*, and *pop*, with teal (greenish) representing lyrical features and amber representing audio features. For *hip hop*, the explanations reveal that lyrical features predominantly drive the model's decision. In contrast, for *punk* music, audio features appear to play a more significant role. For *pop* music, neither audio nor lyrical features dominate, suggesting a more balanced influence from both modalities. These insights, which cannot be fully derived from unimodal explanations due to the lack of direct comparison between feature importances, align with the nature of each genre. *hip hop*'s strong lyrical focus, *punk*'s distinctive musical characteristics, and *pop*'s more diverse thematic content are reflected in the explanations. These findings are further supported by the global lyrical explanations shown in Figure 3. This 2D visualization of the top 10 lyrical features for *hip hop*, *heavy music*, and *pop* reveals that genres where lyrical features dominate also exhibit distinct thematic topics. For instance, *hip hop* features predominantly revolve around street culture, slurs, slang, and artistic expressions, while *heavy music*'s lyrical content centers on dark themes and fantasy elements. Conversely, *pop* music's lyrical content lacks a dominant thematic focus, leading the multimodal model to rely on both audio and lyrics for accurate genre classification.

Our findings, further detailed on our GitHub repository [9], align with the established characteristics of various music genres and asso-

[9]https://github.com/IamTheo2000/MusicLIME

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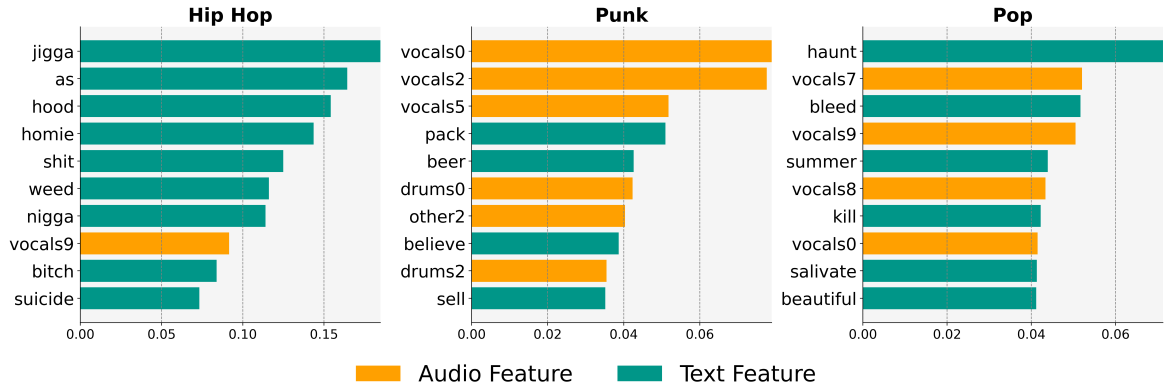

Fig. 2: Top 10 features from the global aggregates for the *hip hop*, *punk*, and *pop* genres from the *Music4All* dataset

ciated emotions, supporting the robustness of our methodology [29]. The multimodal explanations we identified align with the anticipated genre-specific and emotion-specific features. For instance, *folk* music frequently incorporates regional instruments and lyrics that reflect rural life. In contrast, *electronic* music is characterized by the prominence of drums and synthesizers. Similarly, the presence of guitars is a defining feature in *rock* music. Regarding emotion tags, the *tense* emotion appears to be strongly associated with vocal elements, likely due to its connection with the *hip hop* genre. Additionally, positive emotions such as *happy* and *exciting* are often correlated with the use of drums, possibly due to their powerful and dynamic sound.

It is noteworthy that the multimodal explanations produced by MUSICLIME are consistent with the observations and assumptions that a user makes based on the performance metrics outlined in Table I. In music emotion recognition, audio emerges as the dominant modality, as evidenced by the marginal performance improvement when incorporating multimodal inputs and the predominance of audio-based features in the explanations for emotion predictions. This result is in strong agreement with the relevant literature [30]–[32]. Conversely, in genre recognition, both modalities contribute significantly, enhancing overall model performance and yielding explanations that attribute nearly equal importance to each modality.

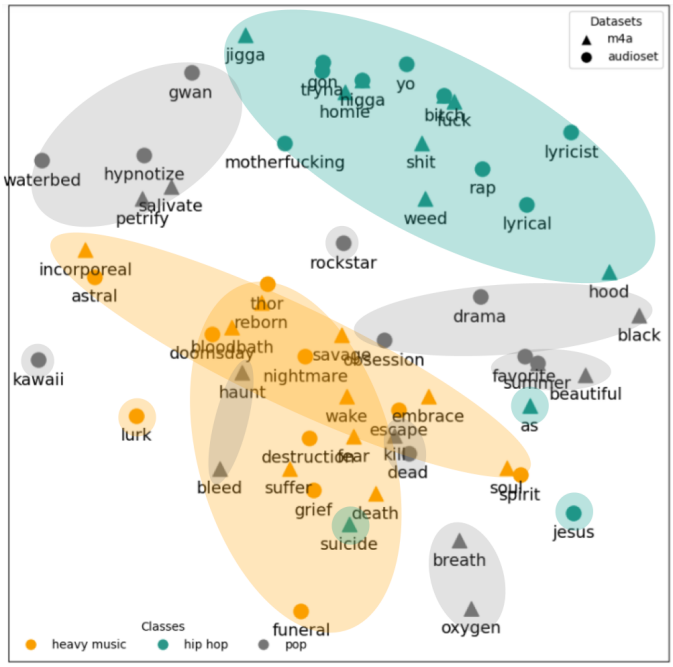

Fig. 3: Top 10 lyrical features for the *heavy music*, *hip hop*, and *pop* genres for both datasets clustered

V. Conclusions & Future Work

In this study, we investigated deep-learning-based multimodal models, curated two multimodal music datasets, and introduced MUSICLIME, a novel, model-agnostic explanation methodology specifically designed for music understanding. Our findings highlight that multimodal approaches outperform unimodal ones by leveraging the complementary information embedded in different modalities and their interactions. We also developed a global aggregation methodology that enhances the interpretation of the relationships between genres or emotions and their associated audio and lyrical features, providing a comprehensive view of the most representative characteristics of each class. We assessed the robustness of MUSICLIME through its application to two distinct datasets and tasks, demonstrating its effectiveness in various contexts.

Future research will focus on enhancing MUSICLIME by improving various pipeline components, including data preprocessing, encoding techniques, and strategies for sample selection and perturbation within the core LIME algorithm. To address the current challenge of defining clear criteria for evaluating the quality of generated explanations, we plan to conduct a human evaluation survey to assess their effectiveness in enhancing music understanding and interpretation. Additionally, since the lyrical modality is currently analyzed at the word level, which may overlook broader contextual meaning, we aim to make MUSICLIME more context-aware to capture more general ideas beyond individual words. Additionally, we will investigate alternative explanation methods, such as counterfactual explanations, and assess their applicability in a multimodal framework for music understanding.

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